

LINKIFIC – HR AUTOMATION PLATFORM

Project README / Technical Project Documentation

1. Project Overview

Linkific is an HR automation platform combining AI-powered interaction with secure Role-Based Access Control (RBAC). It centralizes employee and task operations and provides a natural-language chatbot interface.

2. Objectives

- Centralize HR operations.
- Reduce repetitive manual work.
- Provide AI-powered natural-language interaction.
- Control operations using roles and permissions.
- Keep backend operations secure.

3. Key Features

- Employee and task management.
- AI-powered chatbot.
- Role-Based Access Control.
- Role-Based Chatbot Task Assignment.
- Secure backend operations.

4. Technology Stack

Technology	Purpose
React	Frontend user interface
FastAPI / Python	APIs and backend business logic
PostgreSQL	Application data storage
SQLAlchemy	Database interaction
AI Chatbot / Tools	Natural-language interaction
Docker	Containerized environment

5. System Flow

User → React Frontend → FastAPI Backend → Authentication / RBAC → AI Chatbot / Tools → Business Logic → PostgreSQL

6. AI Chatbot

Users interact with the HR system using natural language. For example: "Show me my tasks." The chatbot identifies the request and calls the appropriate backend functionality.

7. RBAC and Security

The backend verifies the authenticated user's role and permissions before protected operations are executed. Authorization is not dependent only on chatbot behavior.

8. Implemented Feature

Role-Based Chatbot Task Assignment allows an authorized Admin to assign an employee a task through the chatbot.

Admin → Chatbot → Identify Task Request → Check Permission → Validate Employee → Create Task → Confirmation

9. Validation and Error Handling

- Validate authenticated user.
- Check role and permission.
- Validate employee existence.
- Validate task information.
- Handle database/task-creation failures safely.

10. Future Improvements

- More AI-powered HR tools.
- Advanced conversational context.
- Granular permissions.
- Better audit logging.
- Confirmation for sensitive actions.
- Improved monitoring and analytics.

11. Conclusion

Linkific combines AI automation with secure role-based access to make HR operations faster, simpler, and more controlled.